

Discrete Time Control Design of Three Phase PWM Rectifiers

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Abstract

A high bandwidth discrete time controller for three phase PWM rectifiers which optimizes the dc-link stiffness with respect to load variations is introduced. Based on one derived discrete time model of the PWM rectifier the control implementation and design is illustrated. The proposed concept is compared to a conventional design and simulations and experimental results are shown.

Introduction

Due to growing regulations on the injection of current harmonics into the utility grid active rectifiers are becoming a popular choice for the generation of a dc voltage from an ac voltage [1], [2]. In three-phase applications the PWM rectifier based on the B6C topology as shown in Fig. 1 is most frequently utilized because of its simple and widely used circuit topology and standardized control structure.

The common model of the PWM rectifier has been introduced in [3]. It embeds the relevant system dynamics of the rectifier in a second order nonlinear model. To handle the nonlinear system dynamics typically a cascaded control structure with nonlinear state feedback decoupling is used. It consists of a current controller for the control of the ac current and a dc-link voltage controller. For the control of the ac current with zero steady state error either a synchronous frame PI (SFPI) regulator [4] or a proportional resonant (PR) regulator [5] is used. To keep the modeling and control of the ac currents as simple as possible the space vector voltages and currents are often represented by complex numbers as illustrated in [6] and [7]. As a result, the transfer function of these complex state quantities may exhibit complex coefficients and may show different dynamics for space vectors of positive and negative frequency, as derived and explained in [8] and [9]. The dc-link voltage control is commonly realized by a PI regulator [3], whose output is given as a reference to the d-current regulator, thereby manipulating the active power transfer between dc link and the utility grid. In [10] the complex space vector control concept for ac quantities has been combined with the typical dc-link controller.

The dc-link capacitor needs to be sized such that it is capable of buffering the dc-link current harmonics and that the dc-link voltage v_{dc} remains sufficiently unaffected by transient load changes of the dc-link load current i_{load} . For an optimized design of a PWM rectifier the selection of a minimal dc-link capacitance is critical to avoid a waste of installation space and unnecessary cost. Furthermore, a small dc-link capacitance might allow using Metallized Polypropylene Film Capacitors (MPFC) as a real alternative to Aluminum Electrolytic Capacitors. MPFC offer several advantages, e.g., they can be exposed to higher operation temperatures, show less degradation and thus provide a higher reliability [11]. In all applications with tight requirements on the dc-link voltage the minimal required dc-link capacitance is constrained by ability of the dc-link regulator to reject load current variations. It is

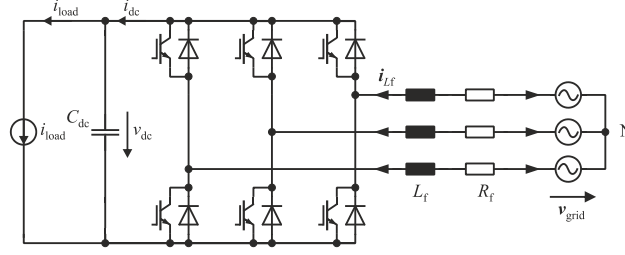


Fig. 1 Topology of the 3 phase PWM rectifier

shown in [12] that a dc-link control structure with higher bandwidth can contribute to a reduction of the dc-link capacitance.

In this work discrete time modeling techniques are used to derive a grid model and a dc-link model of the PWM rectifier. Both models are merged to one discrete time model of the PWM rectifier, which is used for the development of a control topology allowing high bandwidth command tracking of the grid current. That allows designing the voltage state feedback gains up to the theoretical optimum achieving high stiffness and bandwidth. Finally, it is analytically shown what design compromise has to be made with respect to the voltage feedback gains in case of limitations in the current command tracking bandwidth. The presented approach enables either the dc-link stiffness of a given PWM rectifier to be increased to an optimum or the dc-link capacitance to be reduced in comparison to a PWM rectifier which is operated with a conventional controller.

Discrete Time Modeling

In the following the modeling of the PWM rectifier is illustrated to obtain one model describing the discrete time dynamics of the entire system. The PWM voltage in each phase of the rectifier is modulated using space vector modulation [13]. Over one sampling period only the average PWM voltage $\bar{v}_{pwm}^{abc}$ of the converter contributes as manipulated input voltage $\underline{v}_m^{abc}$ to a change of the averaged inductor current $\underline{i}_{Lf}^{abc}$. With this space averaging the higher order harmonics in the converter voltage can be neglected for the averaged system dynamics. Thus, the pulse width modulation process can be simply modeled as a latch [14]. Similarly, the grid voltage with its angular frequency ω_e is included in the model with a latch in the dq-frame in order to capture its sinusoidal characteristics. The resulting discrete time model for the ac side has been derived in [15] and is depicted in Figure 2.

An accurate discrete time model of the dc link is developed in the following. The dynamics of the grid current regulator, which is used to manipulate the dc-link current i_{dc} , have a strong impact on the dc-link dynamics. For the development of the dc-link model it is assumed that dynamics of the grid current can be modeled by a first order lag filter with the time constant τ and a delay of one sampling time T_s , which takes into account the time delay between the sampling of the state variables and the update of the PWM carrier (1).

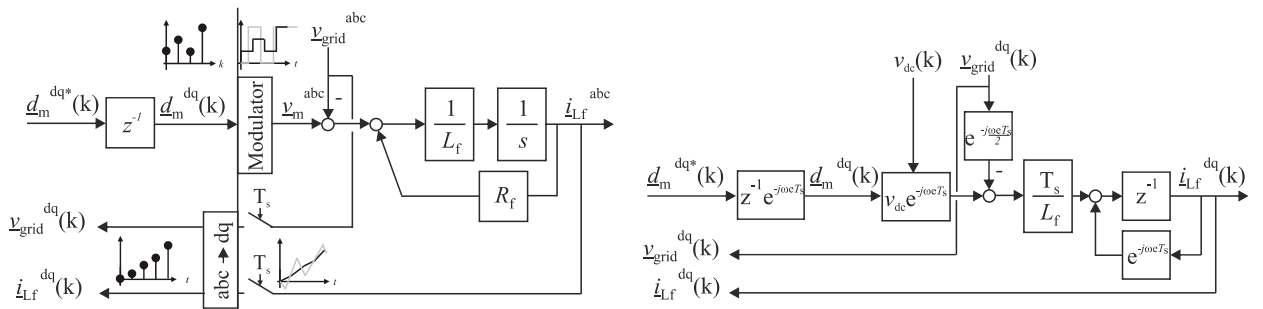


Fig. 2 Full model (left) and discrete time model of the PWM rectifier grid interface (right)

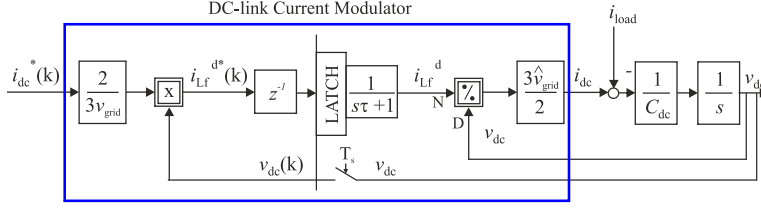


Fig. 3 Detailed model of the dc-link

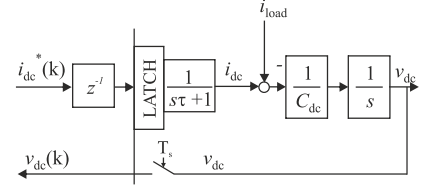


Fig. 4 Simplified dc-link model

$$\frac{I_{Lf}^d(s)}{I_{Lf}^{d*}(s)} = \frac{1}{\tau s + 1} e^{-sT_s} \quad (1)$$

The power balance between the grid side and the dc link $P_{ac} = P_{dc}$ of the converter assuming zero losses provides an expression for the dc-link current i_{dc} as a function of the grid voltage v_{grid}^{dq} , the dc-link voltage v_{dc} and the grid current i_{Lf}^{dq} can be derived according to (2). To simplify this expression the dq frame is oriented such that the grid voltage lies entirely in the d-axis $v_{grid}^d = v_{grid}$ (3).

$$P_{ac} = \frac{3}{2} (v_{grid}^d i_{Lf}^d - v_{grid}^q i_{Lf}^q) = \frac{3}{2} v_{grid}^d i_{Lf}^d \quad (\text{with } v_{grid}^q = 0) \quad P_{dc} = v_{dc} i_{dc} \quad (2)$$

$$i_{dc} = \frac{3}{2} \frac{v_{grid}}{v_{dc}} i_{Lf}^d \quad (3)$$

The dc-link current i_{dc} transferred from the utility grid by the converter and the load current i_{load} are causing the dc-link voltage v_{dc} to change. For the control of the dc-link voltage it is of great importance that the dc-link current can be manipulated accurately. This can be achieved by inverting (2), which results in the dc-link modulator marked in Figure 3. For dc-link voltages which do not vary rapidly it can be assumed that the nonlinear state feedback path of the dc-link voltage decouples the system nonlinearities sufficiently. As a result, for the control design the nonlinear system dynamics from Figure 3 can be simplified to the linear system shown in Figure 4. These linear system dynamics can be used to derive a discrete time model of the dc link. First, the discrete time transfer function of the dc-link current modulator is derived assuming that the current response has an exponential characteristic.

$$\frac{I_{dc}(z)}{I_{dc}^*(z)} = (1 - z^{-1}) Z \left\{ \frac{1}{s} \frac{1}{s\tau + 1} e^{-sT_s} \right\} = \frac{(1 - e^{-T_s/\tau}) z^{-2}}{1 - e^{-T_s/\tau} z^{-1}} \quad (4)$$

Similarly, the discrete time dynamics of the dc-link voltage $v_{dc}(k)$ can be obtained according to (5).

$$\frac{V_{dc}(z)}{I_{dc}^*(z)} = (1 - z^{-1}) Z \left\{ \frac{1}{s^2 C_{dc}} \frac{1}{s\tau + 1} e^{-sT_s} \right\} = \frac{T_s}{C_{dc}} \frac{\mu + (1 - \mu)z^{-1}}{1 - z^{-1}} \frac{(1 - e^{-T_s/\tau}) z^{-2}}{1 - e^{-T_s/\tau} z^{-1}}, \quad \mu = \frac{1}{1 - e^{-T_s/\tau}} - \frac{\tau}{T_s} \quad (5)$$

Both models are merged into one model as shown in (Fig. 5), which exhibits the discrete time dynamics of the entire PWM rectifier. The system parameters of the model are the sampling time T_s , the dc-link capacitance C_{dc} and grid inductance L_f , the grid frequency ω_e and the current loop bandwidth τ .

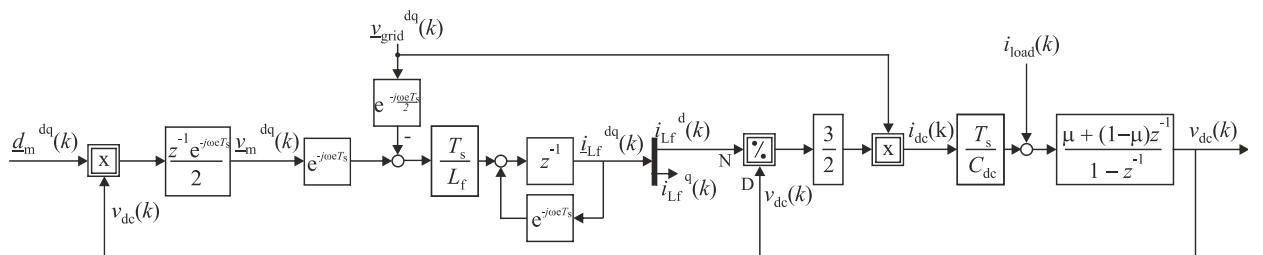


Fig. 5 Discrete time model of the PWM rectifier assuming current command tracking with τ

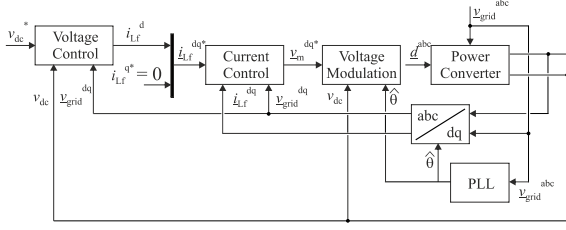


Figure. 6 Synchronous frame control implementation

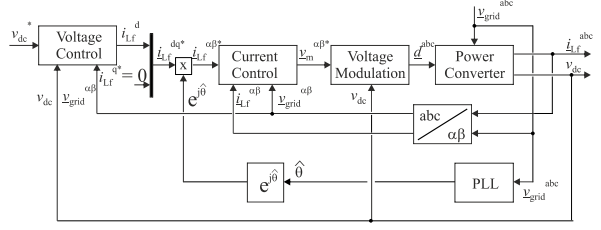


Figure. 7 Stationary frame control implementation

Control design and implementation

The PWM rectifier is operated with a cascaded control structure, which can be either implemented in the synchronous or in the stationary frame (Fig. 6 and 7). The difference between both implementation options is that for current control either all three phase quantities are explicitly transformed into the synchronous frame or they remain in the stationary frame. The voltage controller sets the inductor current that is required in the d-axis i_{Lf}^d in phase with the grid voltage to adjust the dc-link voltage v_{dc} according to its reference v_{dc}^* . The q-axis inductor current reference i_{Lf}^{q*} is set based on the desired reactive power to be fed into the grid, which is typically zero. The current controller calculates the manipulated input voltage v_m^{abc*} required to make the inductor current i_{Lf}^{abc} follow its reference. For the grid synchronization the phase-locked loop (PLL) estimates the angle θ of the grid voltage space vector $v_{grid}^{\alpha\beta}$ to ensure that the dc-link voltage regulator commands an inductor current in phase with the grid voltage resulting in active power flow only. The focus of the following section are the implementation and design details of the voltage and current controller as well as the PLL.

Phase-Locked Loop

The only task of the PLL is the accurate estimation of the grid voltage angle $\hat{\theta}$ to ensure that the injected current has the correct phase shift with respect to the grid voltage such that the desired active and reactive power drawn from the utility grid. An advantageous implementation of a PLL is shown in Fig. 8: The PLL shows an intrinsically nonlinear behavior. To prevent the amplitude of the grid voltage to have an impact on the PLL dynamics, first, the grid voltage $v_{grid}^{\alpha\beta}$ is normalized to a space vector with unity amplitude $e^{\alpha\beta}$. Based on the initial angular velocity estimate $\hat{\omega}_e$ the phase angle estimate is computed with a forward Euler integration. This angle estimate is used to calculate the estimated normalized grid voltage $\hat{e}^{\alpha\beta}$. If the estimate $\hat{e}^{\alpha\beta}$ deviates from the normalized grid voltage $e^{\alpha\beta}$ the cross product of both quantities provides a good measure of the error especially if the deviation is small. The error signal is fed into a PI regulator, which corrects the initial angular velocity estimate $\hat{\omega}_e$ until the angle is estimated correctly.

The PLL compensator parameter K_p^{PLL} and K_i^{PLL} can be designed assuming that the error in the grid

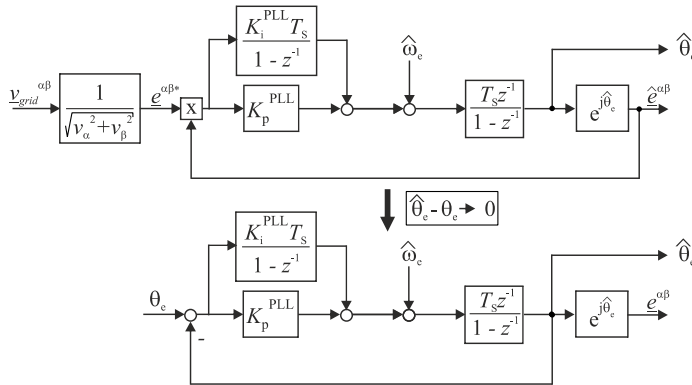


Figure. 8 PLL structure and model for small angle errors

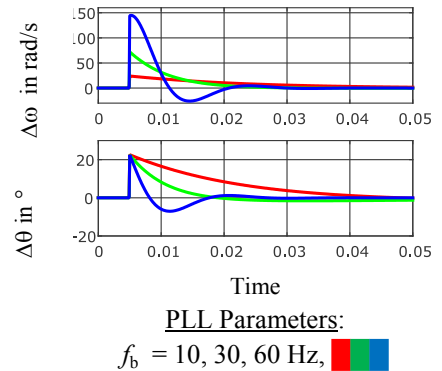


Figure. 9 PLL response to $\Delta\theta = \pi/8$ step

angle estimate is small. In this case the complex exponential function and the cross product can be approximated by the angle difference reducing the nonlinear dynamic of the PLL to linear system dynamics (Fig. 8). The closed loop bandwidth f_b of this feedback loop can be set according to (6). To prevent an interaction of the eigenvalues of both feedback paths the bandwidth of the integral state feedback loop is set one decade below the bandwidth of the proportional feedback loop.

$$K_p^{\text{PLL}} = 2\pi f_b, \quad K_i^{\text{PLL}} = 2\pi \cdot 0.1 f_b \quad (6)$$

It can be seen in Fig. 9 how the PLL responds to a phase jump of $\pi/8$. A bandwidth of 30 Hz seems to provide a reasonable compromise, which allows a fast settling time without overshoot. Note that the normalization allows this PLL to be used independent of the voltage level for any 3-phase application.

Implementation Frame

If the current controller is implemented in the synchronous frame (Fig. 6) based on the angle $\hat{\theta}$ estimated by the PLL all three phase quantities can be transformed into the synchronous frame. Alternatively the controller can be implemented in the stationary frame (Fig. 7). Then the phase information of the grid voltage is used to generate a smooth reference for the current regulator $\hat{i}_{Lf}^{\alpha\beta*}$.

$$\underline{i}_{\text{Lf}}^{\alpha\beta*} = \underline{i}_{\text{Lf}}^{\text{dq}} \cdot \mathbf{e}^{\mathbf{j}\theta(k)} \quad (7)$$

The implementation of the current controller in the synchronous frame is more popular therefore pursued in this paper. However, it is important to understand that the frame, where the regulator is implemented does not have any impact on the system dynamics. The control law in the dq-frame can be transformed into an equivalent control law in the $\alpha\beta$ -frame and vice versa according to (8) [15].

$$z^{-1}|\alpha\beta\rangle \rightarrow z^{-1}e^{-j\omega_e T_s}|\text{dq}\rangle \quad z^{-1}|\text{dq}\rangle \rightarrow z^{-1}e^{j\omega_e T_s}|\alpha\beta\rangle \quad (8)$$

Despite the actual implementation frame the control structure in the stationary frame makes clear that the PLL does not need to have any influence on the dynamics of the current controller. The only parameter of importance for the current controller is the slope of the estimated grid angle $d\theta/dt$ in the synchronous frame or the estimated angular velocity ω_e in the stationary frame: Both can be precomputed as the frequency of the utility grid in most applications does not vary significantly.

However, the PLL has a significant impact on the dc-link voltage regulator. If it does not estimate the grid angle accurately current is injected with a wrong phase angle and thus no active power but reactive power is transferred into the grid. Of course the dc-link regulator will compensate for a certain error, but the effect will be an undesired injection of reactive power in the grid and a higher peak current in the rectifier.

Current Controller

For the control of the grid current $\hat{i}_{L_f}^{\text{dq}}$ a control structure is derived based on the grid interface model (Fig. 2). Based on the manipulated input voltage $\underline{v}_m^{\text{dq}}$ and the dc-link voltage v_{dc} the duty cycle $\underline{a}^{\text{abc}}$ is calculated each sampling period. Thereby, any variations of the dc-link voltage are decoupled and it is ensured that the desired voltseconds are applied by the converter. The state feedback paths $\underline{v}_{\text{dec},i}^{\text{dq}}$ and

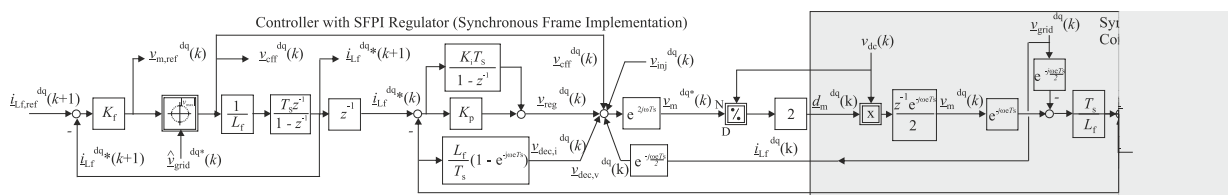


Figure. 10 Discrete time current control of the PWM rectifier in the synchronous reference frame

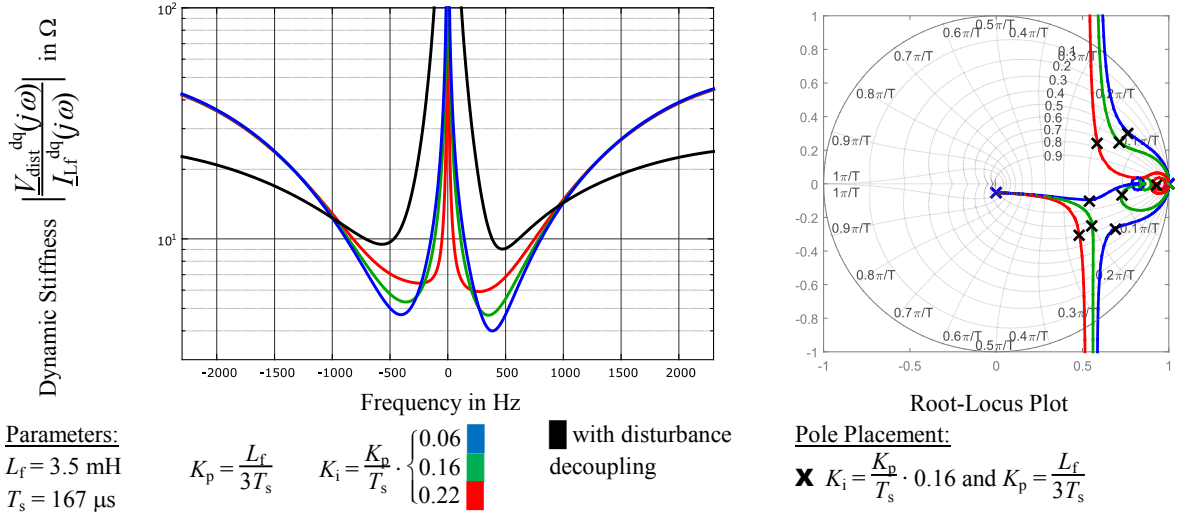


Fig. 11 Root-Locus and Dynamic stiffness plot of the current control loop

$\mathbf{v}_{\text{dec},v}^{\text{dq}}$ decouple variations of the grid voltage as well as the crosscoupling between d- and q-current based on the converter model. The PI regulator in the synchronous reference frame ensures zero steady state error of the fundamental grid current and compensates for all grid disturbances. To optimize the current regulator it is important to increase the dynamic stiffness, which is the disturbance voltage necessary to change the inductor current by 1A, while keeping the system damping ξ in a reasonable range, e.g., $\xi < 1/\sqrt{2}$. To review all design options the PI regulator transfer function can be reshaped to the following expression with the tuning gain $r = K_i T_s / K_p$.

$$\underline{G}_{\text{SFPI}}(z)^{\text{dq}} = K_p + \frac{K_i T_s}{1 - z^{-1}} = K_p \left(1 + \frac{K_i T_s}{K_p} \frac{1}{1 - z^{-1}} \right) \quad (9)$$

Plotting the complex root locus for different tuning gains r (Fig. 11) allows optimizing the regulator design. One simply has to increase the tuning gain until the root locus is about to leave the desired damping range. For this resulting root locus the gain K_p needs to be selected such that the poles are placed exactly at the boundary of the damping range. This is typically the case for $r = 0.16$ with a proportional state feedback gain of $K_p = L_f / (3T_s)$ allowing a standardized regulator design optimized for disturbance rejection for any three phase inductor current control loop. The dynamic stiffness of the system is plotted for different design scenarios in Fig. 11 [15]. It shows that the higher the tuning gain and thus the integral state feedback gain is chosen, the wider the bandwidth of the 0 Hz resonance. As a result the rejection of disturbances at the fundamental frequency is improved. However, if the gain is increased above a certain level undesirable minima arise next to the resonance band. In case the disturbances are measured and decoupled by the disturbances decoupling path, the stiffness is further augmented.

In many applications [3] the current regulator with feedback decoupling paths is used without any

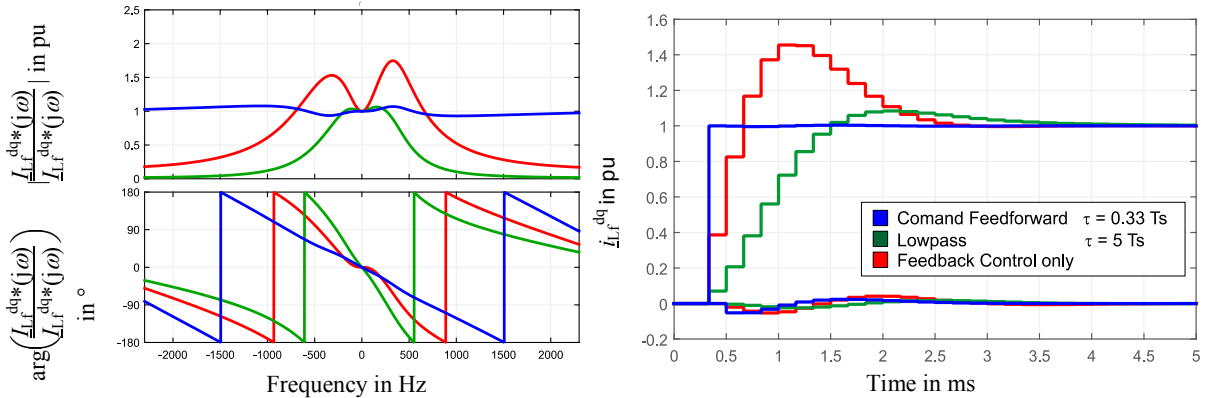


Fig. 12 Frequency response and step response plot for different command filter structures

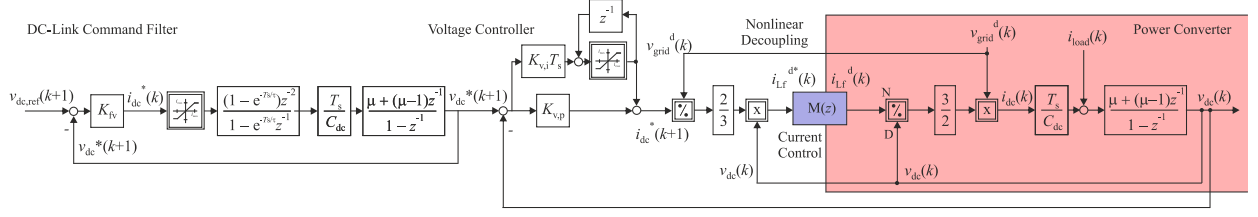


Figure. 13 Discrete time voltage control of the PWM rectifier

further command filter or command feedforward path. In Fig. 12 the command tracking characteristic of the designed feedback controller is depicted for that particular case. The overshoot resulting from a dominant zero in the frequency response function is not tolerable. Note that this does not happen due to low damping. One popular choice to handle this problem is to detune the feedback regulator. However, this is not the best choice as it makes the control system more vulnerable to grid voltage disturbances. Alternatively, a low pass filter can be introduced, which compensates the zero. The result is a smooth command tracking performance with a time constant of $\tau = 5 T_s$. Adding a trajectory filter is an even better option. It computes a feasible reference current i_{Lf}^{dq} and a command feedforward voltage v_{eff}^{dq} for the converter taking into account the limitation of the dc-link voltage and can significantly increase the command tracking bandwidth to achieve deadbeat response properties given that $\exp(-3) \approx 0$ with $\tau = 0.33 T_s$. In the next section it will be illustrated what benefit the increased current command tracking bandwidth brings in the design of the dc-link voltage controller of the PWM rectifier.

Voltage Control

To control the dc-link voltage with high accuracy it is important to decouple the system nonlinearity resulting from the operation point dependent relation between the dc-link current i_{dc} and the d-component of the grid current i_{Lf}^d (3). The dc-link controller decouples this nonlinearity using equation (3) as it is depicted in Fig. 13 such that the dc-link current can be directly modulated with the d-component of the grid current i_{Lf}^d as it has been shown in Fig. 3.

With the dc-link current as manipulated input the PI regulator ensures zero steady state error of the dc-link voltage in the presence of varying load currents. As the grid current and the dc-link current of the converter is limited, it is important to shape a command trajectory which takes into account this limit such that the control system is operated safely. This is realized within the trajectory generation which does not only include a current limit, but also takes into account the limited bandwidth of the current loop and the dynamics of the dc-link capacitor. Furthermore, in case any limit is hit, the accumulation process of the PI regulator is stopped with an Anti-Wind-Up while the limitation is active.

Parameters:

$$C_{dc} = 1.1 \text{ mF}$$

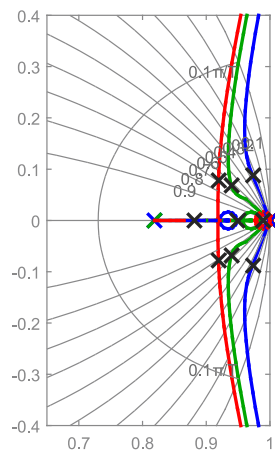
$$T_s = 167 \text{ } \mu\text{s}$$

$$\tau = 0.33 T_s$$

Control Parameters:

$$K_{vp} = 0.3 \frac{C_{dc}}{T_s}$$

$$K_{vi} = \frac{K_{vp}}{T_s} \cdot \begin{cases} 0.10 \\ 0.14 \\ 0.20 \end{cases}$$



Parameters:

$$C_{dc} = 1.1 \text{ mF}$$

$$T_s = 167 \text{ } \mu\text{s}$$

$$\tau = 5 T_s$$

Control Parameters:

$$K_{vp} = 0.07 \frac{C_{dc}}{T_s}$$

$$K_{vi} = \frac{K_{vp}}{T_s} \cdot \begin{cases} 0.01 \\ 0.03 \\ 0.07 \end{cases}$$

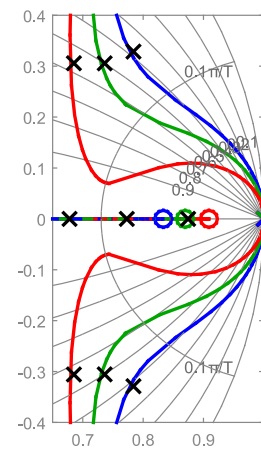


Fig. 14 Root-Locus plots of the voltage control loop for different current modulator bandwidths τ

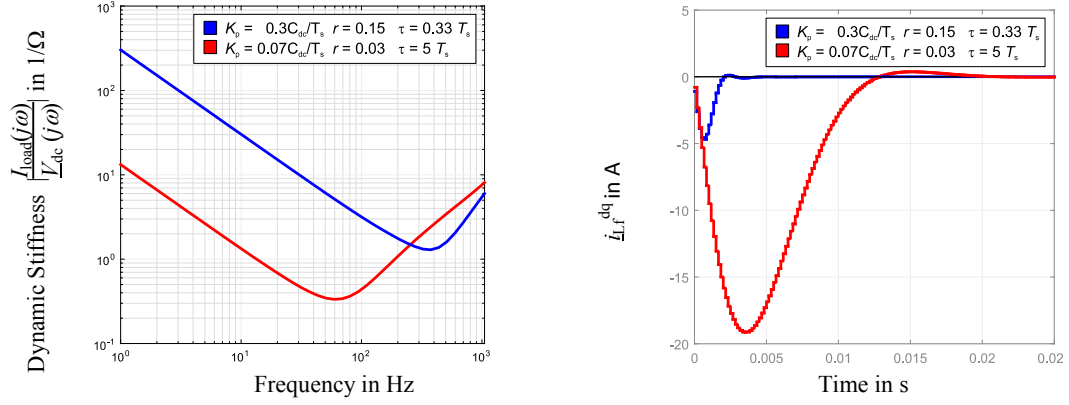


Fig. 15 Dynamic stiffness plot of the voltage control loop and disturbance response to a 10 A load step with the following system setup $C_{dc} = 1.1 \text{ mF}$, $T_s = 167 \text{ } \mu\text{s}$, $\tau = 3 T_s$

For the design of the voltage regulator gains the model of the dc-link and the PI regulator (10) are used. The transfer function of the PI regulator is reshaped and a tuning gain $r = K_{vi}T_s/K_{vp}$ is introduced.

$$G_{plant}(z) = \frac{T_s}{C_{dc}} \frac{\mu + (1 - \mu)z^{-1}}{1 - z^{-1}} \frac{(1 - e^{-T_s/\tau})z^{-2}}{1 - e^{-T_s/\tau}z^{-1}} \quad \mu = f\left(\frac{T_s}{\tau}\right) \quad G_{PI}(z) = K_{vp} \left(1 + \frac{K_{vi}T_s}{K_p} \frac{1}{1 - z^{-1}}\right) \quad (10)$$

Optimized feedback gains can be identified following the identical procedure as it was used for the grid current regulator design. The tuning gain is increased until the root locus leaves the desired damping range $\xi < 1/\sqrt{2}$ and for the resulting tuning gain the proportional and integral feedback gain are selected. However, the command tracking bandwidth of the current regulator τ has a strong impact on this design process. A higher command tracking bandwidth of the current regulator allows the voltage regulator feedback gains to be designed much higher. In Fig. 14 the design procedure for the voltage regulator is illustrated for a current controller with deadbeat command tracking properties $\tau = 0.33 T_s$ (right) and for a bandwidth of a state of the art regulator design with a simple low-pass as a pre-filter. The dynamic stiffness and disturbance response of the dc-link voltage resulting from both designs, shown in Fig. 15, demonstrate the huge difference the command tracking bandwidth of the current regulator has on the tuning options of the voltage loop.

The feedback gains can be physically interpreted in a way that the worst case stiffness is nearly equal to the proportional voltage feedback gain. This property also determines the voltage sag as result of a certain load step $|\Delta v_{dc}| = |i_{Load}/K_{vp}|$. The low frequency stiffness resulting in the settling time after a load step can be manipulated by the integral feedback gain.

To determine the optimal feedback gains of the voltage regulator given the command tracking time constant of the current regulator one can follow the illustrated design approach for various ratios of λ

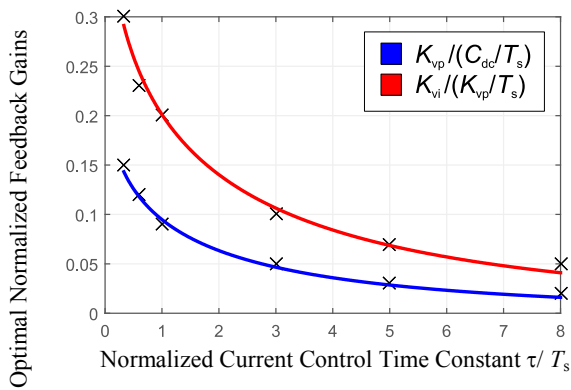


Fig. 16 Optimization of the voltage controller

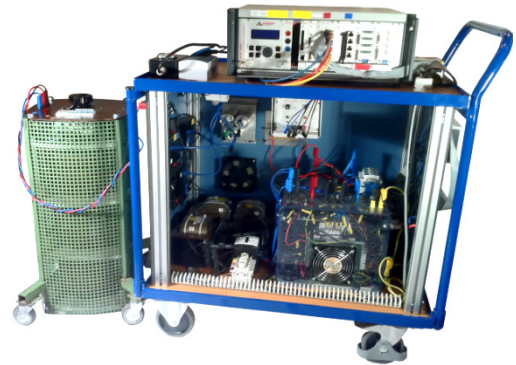
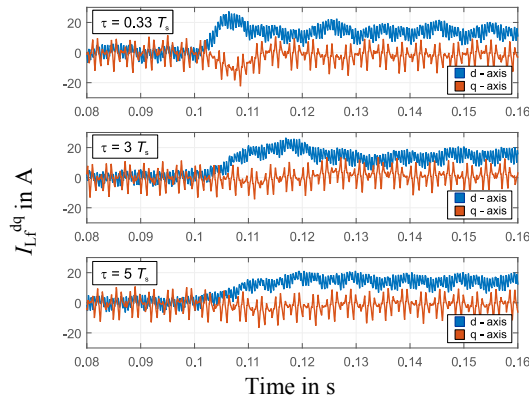
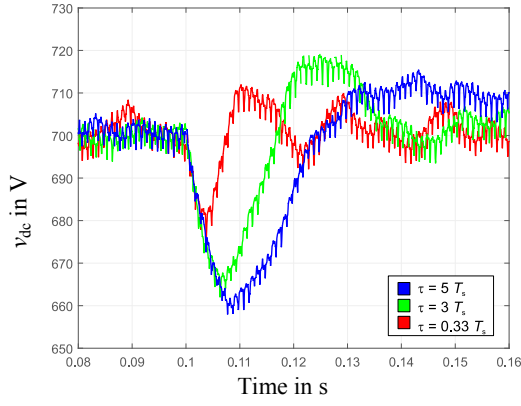


Figure 17 Experimental setup of the PWM rectifier



Parameters

$$L_f = 3.2 \text{ mH}$$

$$C_{dc} = 1.1 \text{ mF}$$

$$R_{load} = 70 \Omega$$

$$f_{pwm} = 1 \text{ kHz}$$

$$f_s = 2 \text{ kHz}$$

$$V_{grid} = 326 \text{ V}$$

Current Control

$$K_p = 0.33 L_f / T_s$$

$$K_i = 0.15 K_p / T_s$$

Voltage Control

According to

Fig. 16

Fig. 18 Experiment: Load current response of the optimized voltage controller for different current command tracking bandwidths τ established in the command filter - Shopped load resistance $R_{load} = 70 \Omega$

(11). Based on the dc-link model (10) it can be shown that the relationship given by equation (11) provides feedback gains optimizing the stiffness of the dc-link given a certain sampling time T_s and a dc-link capacitance C_{dc} . This relationship is illustrated in Fig. 16.

$$K_{vp} = \frac{C_{dc}}{T_s} 0.48 e^{-0.87\sqrt{\lambda}} \quad K_{vi} = \frac{K_{vp}}{T_s} 0.25 e^{-0.97\sqrt{\lambda}} \quad \lambda = \tau/T_s \quad (11)$$

The optimization of the disturbance response properties allows either to design a given hardware with improved properties or it allows reducing the size of the dc-link capacitor. However, in case of bandwidth limitations of the dc-link voltage measurement due to the sensors used or noise injections a control design with reduced disturbance rejection ability might be preferable to prevent that the regulator gets sensitive to that high frequency range.

Experimental Results

To validate the introduced control topology and design methodology the control structure was implemented in C++ on an AixControl XCS2000 DSP system and the control algorithms were tested on the setup shown in Fig.17 with the parameters listed in Fig. 18. The dc-link voltage was controlled to 700 V when a load resistance of 70Ω was connected to the dc-link. The resulting voltage sag and

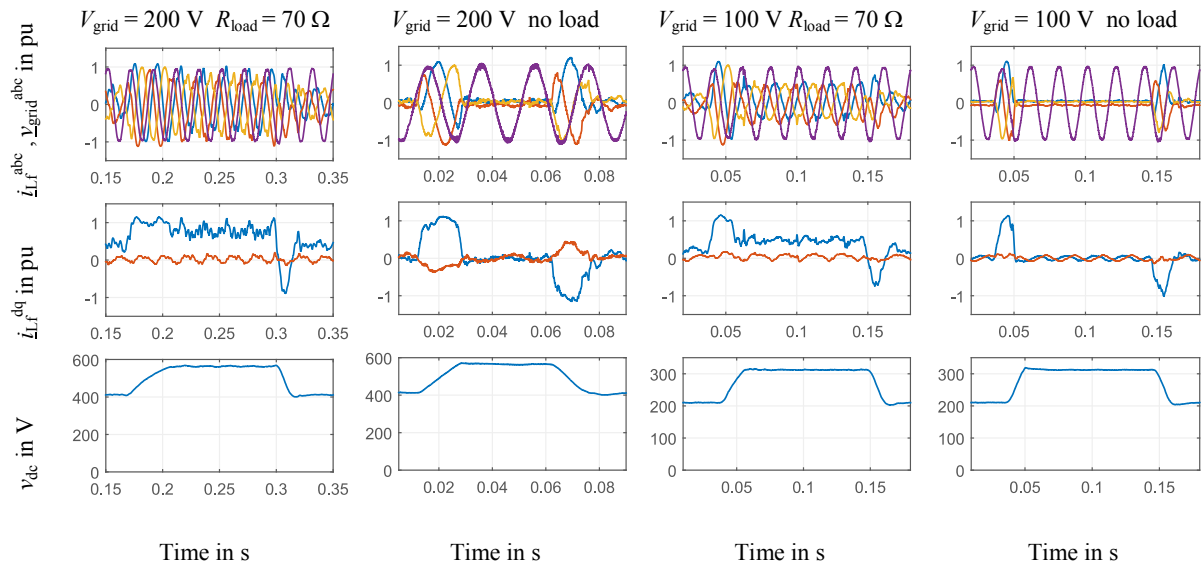


Fig. 19 Experiment: DC-Link Voltage Command Tracking at different grid voltages with and without load

Parameters according to Fig. 18 but with $f_{pwm} = 3 \text{ kHz}$, $f_s = 6 \text{ kHz}$ and $I_r = 20 \text{ A}$

the corresponding currents in the dq-frame are shown in Fig. 18. It can be seen that the higher the current loop bandwidth $1/\tau$ the larger the dc-link regulator gains can be sized and thus the voltage sag can be reduced for a given load current variation. The regulator structure also allows fast changes of the dc-link voltage. For the experiment shown in Fig. 19 the grid voltage was reduced and the PWM and sampling frequency were increased by a factor of 3. The results show that fast changes of the dc-link voltage are possible without overshoot and that the current limits included in the controller and the anti-windup prevents the rated current to be exceeded. Furthermore, this second experiment demonstrates that the calculation of the feedback gains and the decoupling of the nonlinearity works for arbitrary voltages and sampling rates.

Conclusions

In this work modeling, implementation and design aspects for three-phase PWM rectifiers and their control have been studied. A control implementation format has been proposed which allows optimizing the bandwidth and the dynamic stiffness of the dc-link voltage regulator. This boost of performance achieved by high bandwidth current command tracking allows designing the PWM rectifiers with smaller dc-link capacitance than required when using conventional control algorithms for given load dynamics. Experimental results demonstrate the validity of the proposed concept.

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